Drift Theorem for Self-Referential Vision Systems

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Abstract

This theorem formalizes drift measurement in a computer vision system where a camera estimates motion relative to observed objects using vector displacement. Rather than tracking absolute object motion, the system evaluates **relative vector change between observer and target**, producing radial and angular drift metrics. The theorem provides a mathematical basis for spatial tracking in OpenCV-based systems and cybernetic sensing architectures.

Definitions

Let:

- (C(t)) = position vector of the camera in 3D space
- (P(t)) = position vector of the observed object
- (V(t)) = relative observation vector

[
V(t) = P(t) - C(t)
]

The magnitude of this vector gives the **observer-object distance**:

[
d(t) = ||V(t)||
]

Drift Theorem

The **drift of an observed object relative to an observer** is defined as the time derivative of the relative observation vector.

$$
D(t) = \frac{dV(t)}{dt}
$$

Where:

- $D(t)$ represents the **drift vector**
- It measures how the spatial relationship between observer and object changes over time.

Radial Drift

Radial drift measures **approach or separation** between the observer and the object.

$$
D_r = \frac{dd(t)}{dt}
$$

Interpretation:

Condition	Meaning
$D_r > 0$	Object moving away
$D_r < 0$	Object approaching
$D_r = 0$	Distance unchanged

Angular Drift

Angular drift measures **lateral displacement**.

Let θ represent the angle between successive observation vectors:

$$
\theta = \cos^{-1}\left(\frac{V(t)\cdot V(t-\Delta t)}{|V(t)||V(t-\Delta t)|}\right)
$$

Angular drift:

$$
\omega = \frac{d\theta}{dt}
$$

Where:

- ω represents **rotational change of the observation vector**.

Weighted Observation Function

To stabilize tracking, define a weighted observation estimate:

$$
W(t) = \alpha P(t) + (1-\alpha)P(t-\Delta t)
$$

Where:

$$
0 < \alpha < 1
$$

Example used in your system:

$$
W(t) = \frac{2}{3}P(t) + \frac{1}{3}P(t-\Delta t)
$$

This reduces noise in frame-to-frame tracking.

Drift State Classification

The observation system classifies motion into discrete states.

State	Condition
Approach	$D_r < 0$

Separation	($D_r > 0$)
Angular Drift	($\omega > 0$)
Stable	($D_r \approx 0$) and ($\omega \approx 0$)

These states may be represented using **color encodings** in a visualization system.

Implementation Principle

In a computer vision framework (such as **OpenCV** with **NumPy**), the drift vector can be computed using centroid tracking between consecutive frames.

$$
V_t = P_t - P_{t-1}
$$

$$
D = ||V_t||
$$

This allows the system to estimate motion using only frame displacement.

Theorem Statement

Drift Theorem

> In a spatial observation system, the motion of a tracked object relative to the observer can be completely characterized by the time derivative of the relative observation vector, producing radial and angular drift components that describe separation, approach, and lateral displacement within the observation frame.

Implication for OBINexus Systems

This theorem allows a vision system to:

- determine motion **without machine learning**
- track objects using **pure vector mathematics**
- operate as a **self-referential spatial observer**

This aligns strongly with cybernetic sensing principles in the field of **Cybernetics**.